

SYNCHRONOUS BALANCED REFINEMENT OF COMPOSITIONS: CANONICAL KERNELS AND EVERY ITERATED FIBRE

ANONYMOUS

ABSTRACT. We iterate the synchronous part rule $m \mapsto (\lfloor m/2 \rfloor, \lceil m/2 \rceil)$ for $m > 1$ on integer compositions. For every iterate t , with $K = 2^t$, we identify the complete kernel congruence: replace each source part at most K by that many ones and retain every larger part. Two compositions have the same t -th image exactly when these canonical forms agree. The proof exposes a suffix code and gives a right-to-left image decoder. Consequently every nonempty t -step fibre is a product of K -step Fibonacci counts over the one-runs of its canonical source, and the largest fibre on weight n is the K -restricted composition number. The same number counts states of depth at most t . Finally, the entire t -th image is in bijection with compositions whose parts lie in $\{1\} \cup \{K+1, K+2, \dots\}$, yielding an explicit rational generating function. Restricted-composition enumeration, suffix-code theory, and balanced divide-and-conquer splitting are treated as background. The owner search is bounded; no novelty, priority, or external-release claim is made.

1. THE MAP AND THE SUBTRACTION BOUNDARY

Let $\mathcal{C}_n$ be the set of integer compositions of n , with the empty composition included when $n = 0$. Define a substitution on positive integers by

$$(1) \quad \sigma(1) = 1, \quad \sigma(m) = (\lfloor m/2 \rfloor, \lceil m/2 \rceil) \quad (m > 1),$$

and extend it by concatenation. Our dynamical map is $\Phi(a_1 \cdots a_r) = \sigma(a_1) \cdots \sigma(a_r)$. All parts are read from the old word, so the update is synchronous. Weight is preserved and every changed part raises the word length; hence 1^n is the unique recurrent state and there are no nontrivial cycles.

Balanced splitting and its divide-and-conquer clock are standard. Free monoid substitutions and uniquely decipherable codes are classical [1]. Enumeration of compositions with parts in a set, including bounded parts, is also established [3, 4]. In particular, the one-step image count below is the no-part-2 class studied by Chinn and Heubach [2]. All of those interfaces, and every generating function obtained merely by choosing an allowed part set, receive zero contribution credit.

The residual question is dynamical and all-iterate: can one identify the kernel of Φ^t , decode its image, and count every individual fibre? The exact rule was subjected to a bounded owner search across rewriting, fragmentation, substitution, and composition sources. No direct theorem package was found; a bounded non-hit is not a novelty certificate.

2. CODEWORDS AND THE EXACT CLOCK

Put

$$W_t(m) = \Phi^t((m)), \quad K = 2^t.$$

These are words, not unordered fragmentation profiles.

Lemma 2.1 (balanced codewords). *For $m \geq 1$, $W_t(m)$ has length $\min(m, K)$, total weight m , and largest and final part $\lceil m/K \rceil$. In particular,*

$$(2) \quad W_t(m) = 1^m \iff m \leq K.$$

Proof. At each split the two children differ by at most one. Induction on t therefore gives largest descendant $\lceil m/2^t \rceil$; following the right child at every split gives this value as the final part. If $m \leq K$, every branch reaches one and weight preservation gives m leaves. If $m > K$, every node before level t is at least two, so all K leaves are present. This proves every assertion. $\square$

For a nonempty composition a , let $M(a) = \max_i a_i$.

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Proposition 2.2 (pointwise and global clocks).

$$\text{depth}(a) = \lceil \log_2 M(a) \rceil, \quad \max_{a \in \mathcal{C}_n} \text{depth}(a) = \lceil \log_2 n \rceil \quad (n \geq 1).$$

The empty composition has depth zero, and (n) attains the global bound.

Proof. By lemma 2.1, after t rounds every descendant is one exactly when every source part is at most 2^t . Taking the first such t gives the pointwise formula. Since $M(a) \leq n$, the global statement and its witness follow. $\square$

Let $R_K(r)$ count compositions of r with all parts at most K , with $R_K(0) = 1$. Thus

$$(3) \quad R_K(r) = \sum_{j=1}^{\min(K,r)} R_K(r-j), \quad \sum_{r \geq 0} R_K(r) x^r = \frac{1}{1 - x - x^2 - \dots - x^K}.$$

The recurrence is the usual K -step Fibonacci convention.

Corollary 2.3 (all depth layers). *The number of states in $\mathcal{C}_n$ of depth at most t is $R_{2^t}(n)$. For $t \geq 1$, the exact depth- t layer has size $R_{2^t}(n) - R_{2^{t-1}}(n)$.*

Proof. Equation (2.2) says precisely that every part is at most 2^t . The enumeration in (3) is classical; only its identification with the temporal layer is used here. $\square$

3. A CANONICAL FORM FOR THE COMPLETE ITERATED KERNEL

For $K = 2^t$, define a weight-preserving monoid map

$$(4) \quad \mathcal{N}_K(m) = \begin{cases} 1^m, & m \leq K, \\ m, & m > K, \end{cases} \quad \mathcal{N}_K(a_1 \cdots a_r) = \mathcal{N}_K(a_1) \cdots \mathcal{N}_K(a_r).$$

Its values are precisely the compositions over the canonical alphabet $\{1\} \cup \{K+1, K+2, \dots\}$.

Theorem 3.1 (canonical kernel and suffix decoder). *For all compositions a, b of the same weight,*

$$(5) \quad \Phi^t(a) = \Phi^t(b) \iff \mathcal{N}_K(a) = \mathcal{N}_K(b).$$

Moreover, Φ^t is injective on canonical forms. Its image code is

$$(6) \quad \{1\} \cup \{W_t(m) : m > K\},$$

which is a suffix code. Hence membership in $\text{im } \Phi^t$ and the unique canonical source are decided by a right-to-left decoder.

Proof. Equation (2) gives $\Phi^t(a) = \Phi^t(\mathcal{N}_K(a))$. It remains to prove injectivity on canonical forms. The codeword for the canonical letter 1 is the one-letter word 1. Every $W_t(m)$ with $m > K$ has length K and, by lemma 2.1, ends in $\lceil m/K \rceil > 1$. Distinct long codewords have distinct weights. Thus no word in (6) is a proper suffix of another.

Explicitly, if a target ends in 1, its final canonical letter must be 1; delete that symbol. Otherwise its final codeword must be the last K symbols. Their sum determines the sole possible source letter $m > K$, and the block is accepted exactly when it equals $W_t(m)$. Delete the block and repeat. This proves unique decoding and injectivity. Applying it to $\Phi^t(\mathcal{N}_K(a)) = \Phi^t(\mathcal{N}_K(b))$ proves the converse in (5). $\square$

For example, take $t = 2$, so $K = 4$. The source $(3, 5, 2)$ has canonical form

$$(1, 1, 1, 5, 1, 1)$$

and image 111111211. Reading from the right deletes two terminal ones, recognizes the block 1112 = $W_2(5)$, and then deletes three ones, recovering the canonical form.

4. EVERY FIBRE AND THE CLOCK-FIBRE DUALITY

Write a canonical form uniquely as

$$(7) \quad z = 1^{r_0} m_1 1^{r_1} m_2 \cdots m_s 1^{r_s}, \quad m_i > K, \quad r_i \geq 0.$$

Theorem 4.1 (complete fibre product). *A target y has no t -step preimage exactly when the suffix decoder in [theorem 3.1](#) rejects it. If the decoder returns the canonical form (7), then*

$$(8) \quad |(\Phi^t)^{-1}(y)| = \prod_{j=0}^s R_K(r_j).$$

Consequently,

$$(9) \quad \max_{y \in \mathcal{C}_n} |(\Phi^t)^{-1}(y)| = R_K(n),$$

attained by $y = 1^n$. Thus the maximum t -fibre equals the number of states of depth at most t .

Proof. By [theorem 3.1](#), the fibre over y is exactly the inverse image under $\mathcal{N}_K$ of its decoded canonical form. Every large letter m_i must remain the same source part. Independently, a run 1^{r_j} may be grouped into any composition of r_j with parts at most K , giving (8).

Concatenation injects a tuple of restricted compositions of lengths $r_0, \dots, r_s$ into a restricted composition of their total length, so the product in (8) is at most $R_K(\sum r_j) \leq R_K(n)$. The final inequality follows by appending ones. For $y = 1^n$, the canonical form is 1^n and equality holds. The last claim now follows from [corollary 2.3](#). $\square$

In the displayed $K = 4$ example, the canonical one-runs have lengths three and two. Hence the image 111111211 has $R_4(3)R_4(2) = 4 \cdot 2 = 8$ two-step preimages. This is pointwise information: different targets of the same weight generally have different products.

5. THE COMPLETE IMAGE TOWER

Let $I_{n,t} = |\text{im}(\Phi^t : \mathcal{C}_n \rightarrow \mathcal{C}_n)|$.

Theorem 5.1 (all-iterate image census). *For $K = 2^t$, restriction of Φ^t is a weight-preserving bijection from canonical forms to its image. Therefore*

$$(10) \quad \sum_{n \geq 0} I_{n,t} x^n = \frac{1}{1 - \left(x + \frac{x^{K+1}}{1-x}\right)} = \frac{1-x}{1-2x+x^2-x^{K+1}}.$$

Equivalently, $I_{0,t} = I_{1,t} = 1$ and, for $n \geq 2$,

$$(11) \quad I_{n,t} = 2I_{n-1,t} - I_{n-2,t} + \mathbf{1}_{n \geq K+1} I_{n-K-1,t}.$$

For $n \geq 1$, the number of Garden states at time t is $2^{n-1} - I_{n,t}$.

Proof. Every source has the same image as its canonical form by [theorem 3.1](#); distinct canonical forms have distinct images. Hence the image is equinumerous with sequences of parts in $\{1\} \cup \{K+1, K+2, \dots\}$. The ordinary sequence construction gives (10); coefficient extraction gives (11). There are 2^{n-1} compositions of positive n , proving the last claim. $\square$

For $t = 1$, the image numbers begin 1, 1, 1, 2, 4, 7, 12, 21, 37, 65, $\dots$; the equinumerous canonical class is exactly the classical compositions without the part 2 [2]. This enumerative class is zero-credit. The result here is the all- t dynamical bijection supplied by the canonical kernel and suffix decoder. When $2^t \geq n$, only 1^n is canonical, so $I_{n,t} = 1$, in agreement with the sharp clock.

6. EXACT CONTROLS, LIMITS, AND CONCLUSION

The paper-local verifier uses only exact integer arithmetic and the Python standard library. It exhausts all compositions through weight 18 for the clock, and all sources and targets through weight 15 for every $0 \leq t \leq 5$. It compares literal kernel classes, suffix decoding, the fibre product, an independent factorization DP, maximum fibres, and the image census. Codewords are checked through $m = 256, t = 8$, and the rational image recurrence through weight 90, $t = 8$. These are falsification controls; the quantified statements rest on the proofs above.

The scope is deliberately finite and exact. We do not claim a new general theory of substitutions, codes, restricted compositions, fragmentation, or balanced trees. We do not treat asynchronous splits, other arities, random schedules, or asymptotic laws. The exact-map owner search remains bounded, so novelty, priority, and external release remain on hold.

Within that boundary, the normal form (4) gives more than a terminal clock: it determines the complete kernel of every iterate. The suffix code turns that kernel into a pointwise image test, the one-runs give every fibre, and the canonical alphabet gives the full image tower. The coincidence between the largest inverse fibre and the cumulative depth census is the resulting clock-fibre duality.

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